

Local AI for confidential subsurface data

The stack we are building at GEUS

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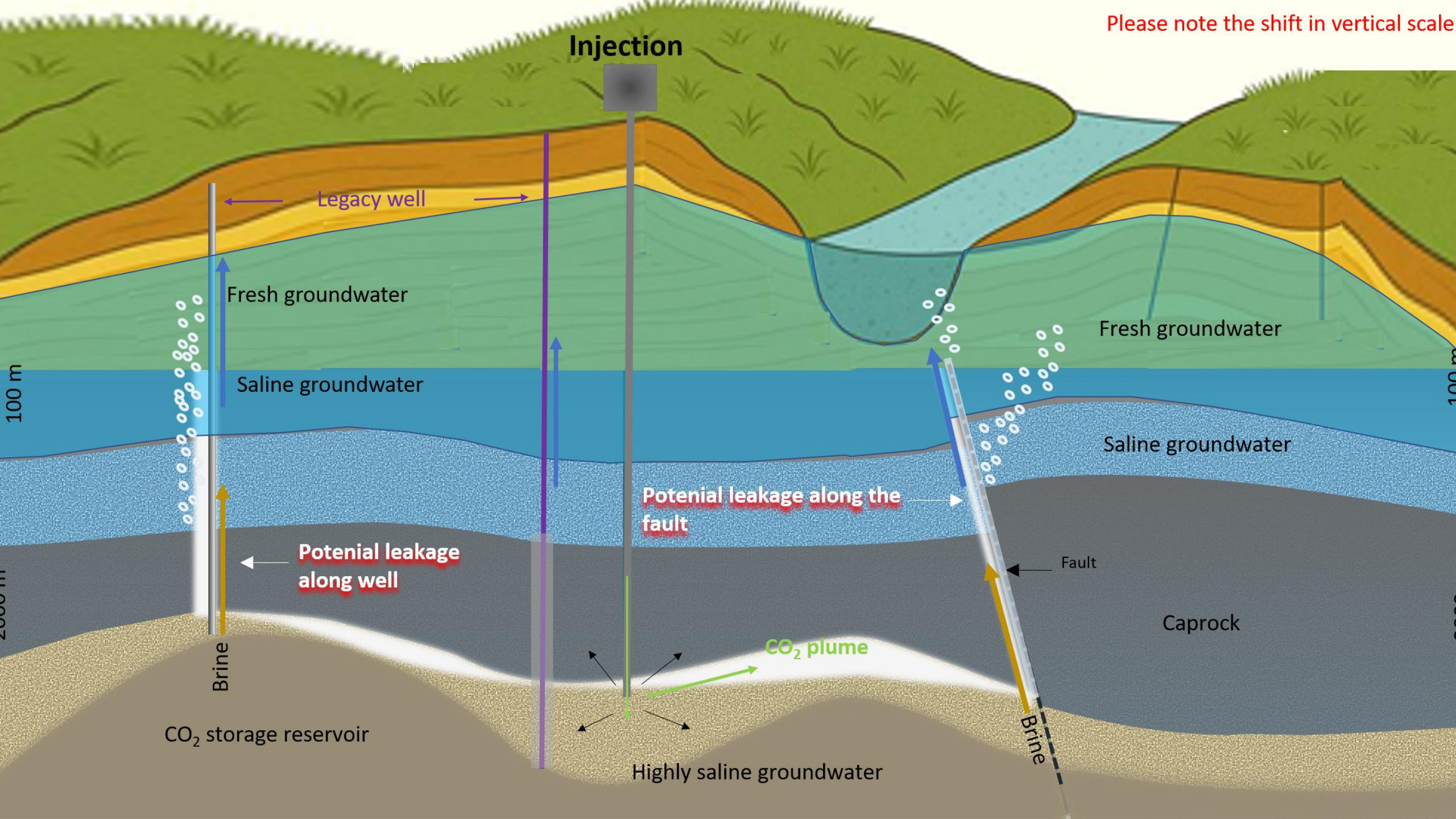
This work has received funding from INNO-CCUS through the OMGCCS project
(**O**ptimized **M**onitoring of **G**round- and surface-water for **C**arbon **C**apture and **S**torage)

and

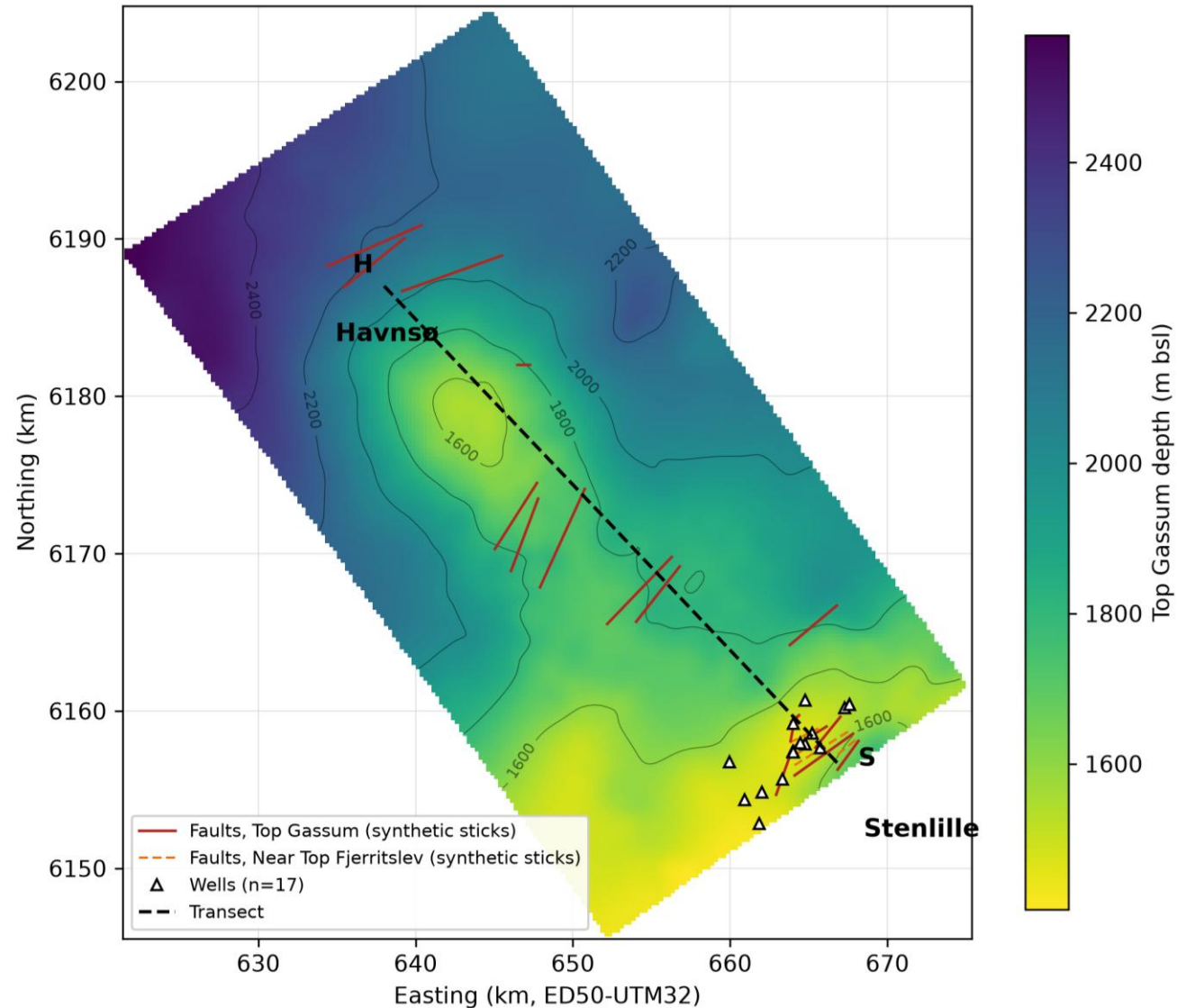
GREENMIN

from the Ministry of Mineral Resources of Greenland through the project
“Exploring AI for Data Mining and Knowledge Extraction”

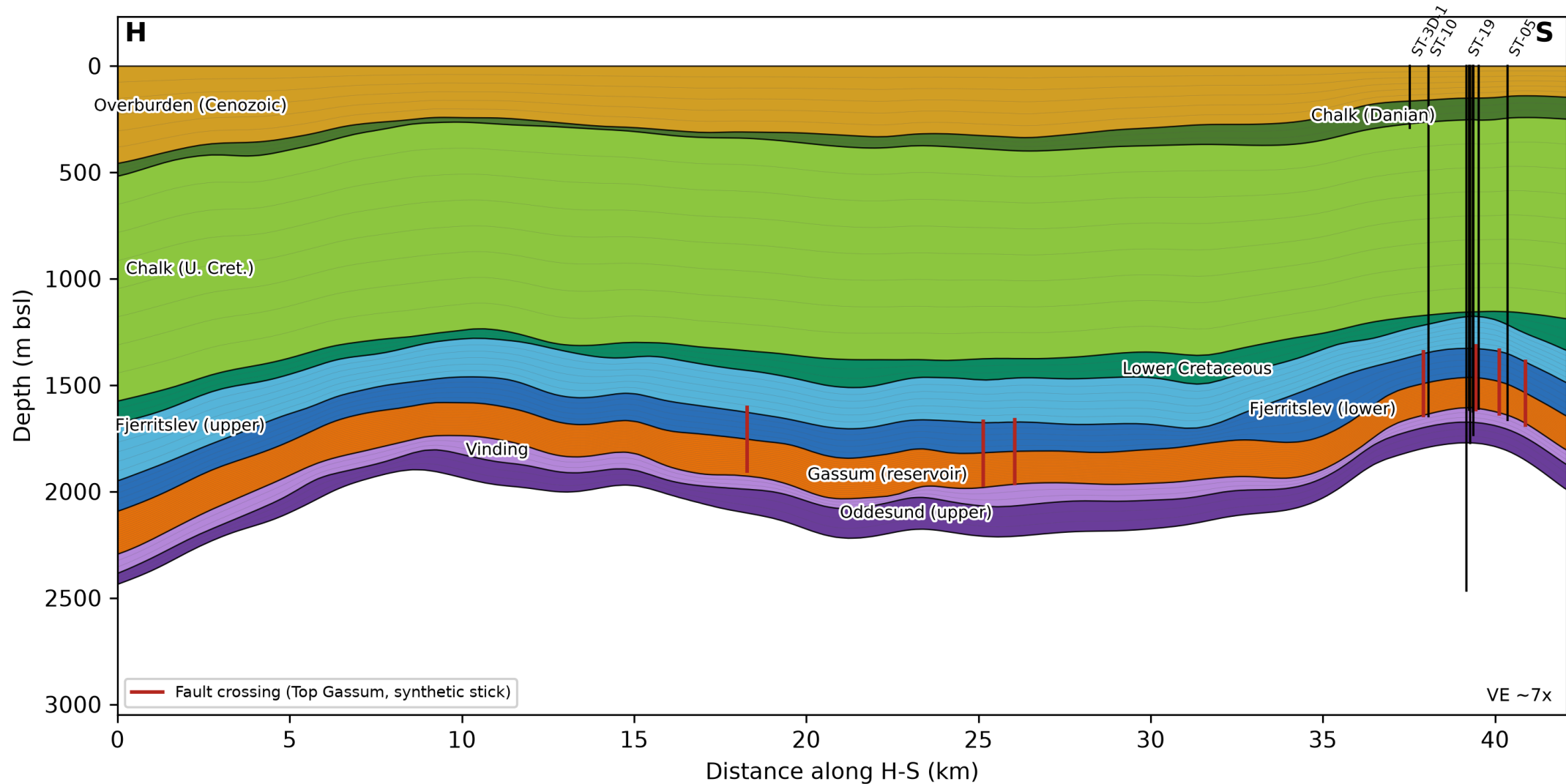
Please note the shift in vertical scale



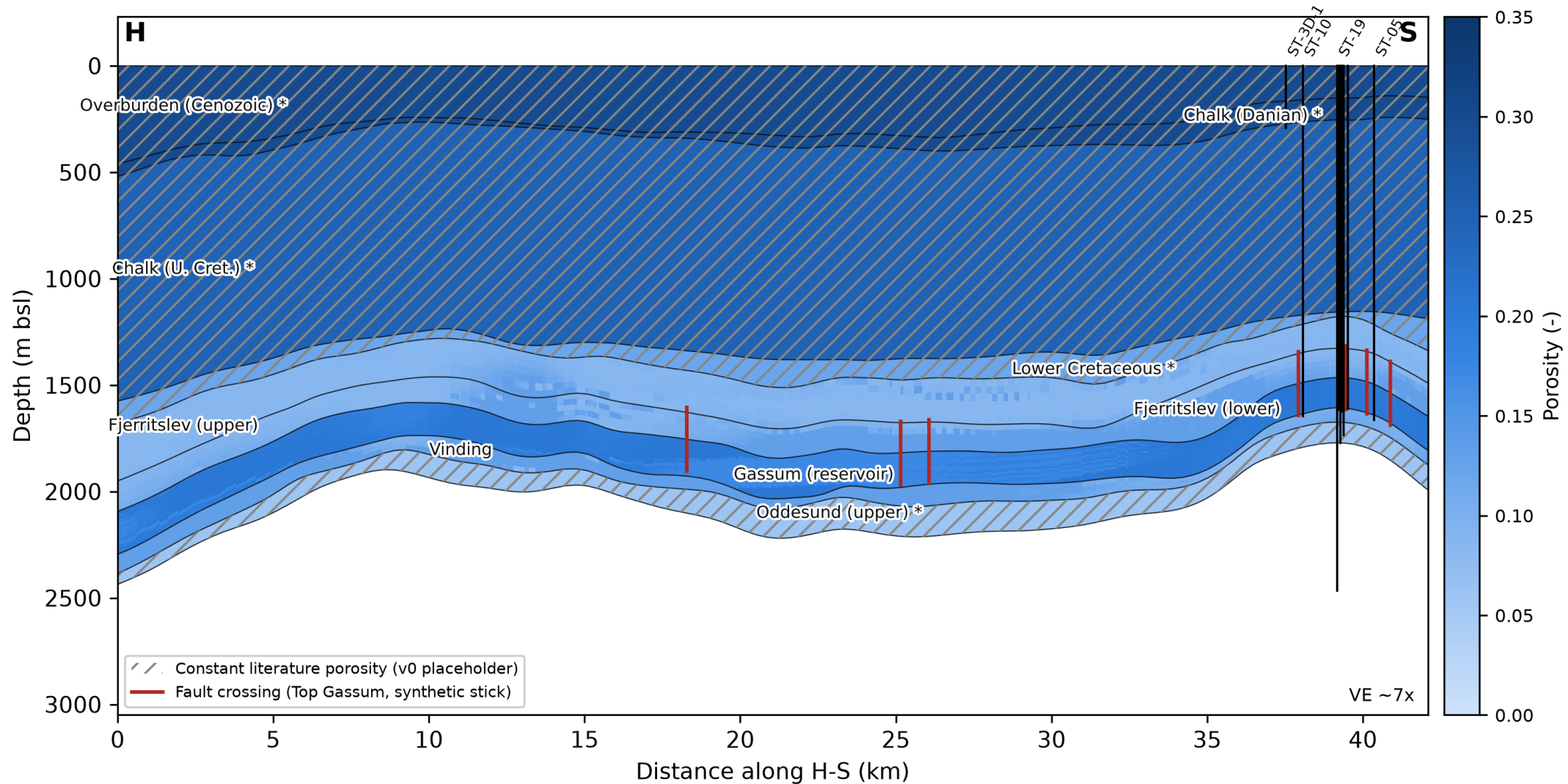
Towards a structural model – Top Gassum



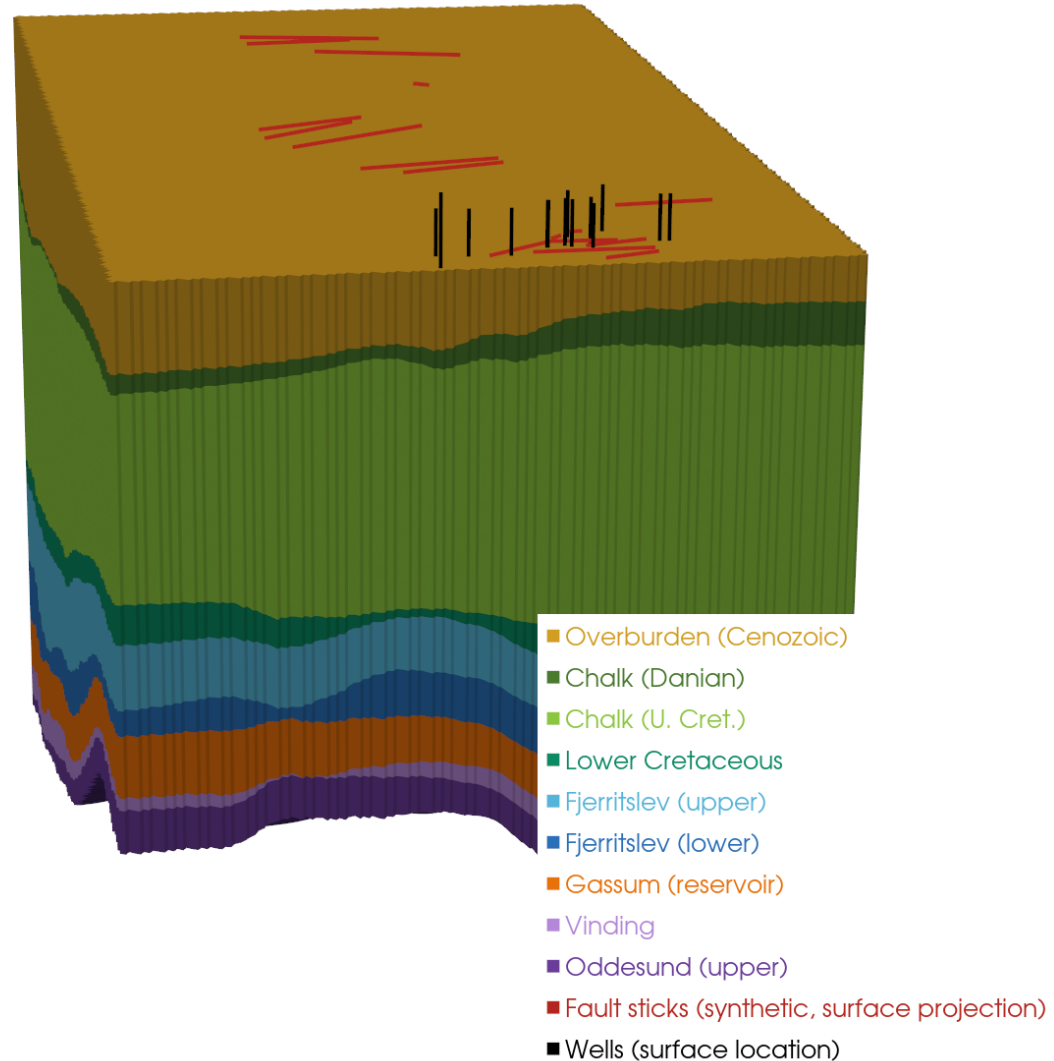
Towards a structural model – Zones along HS



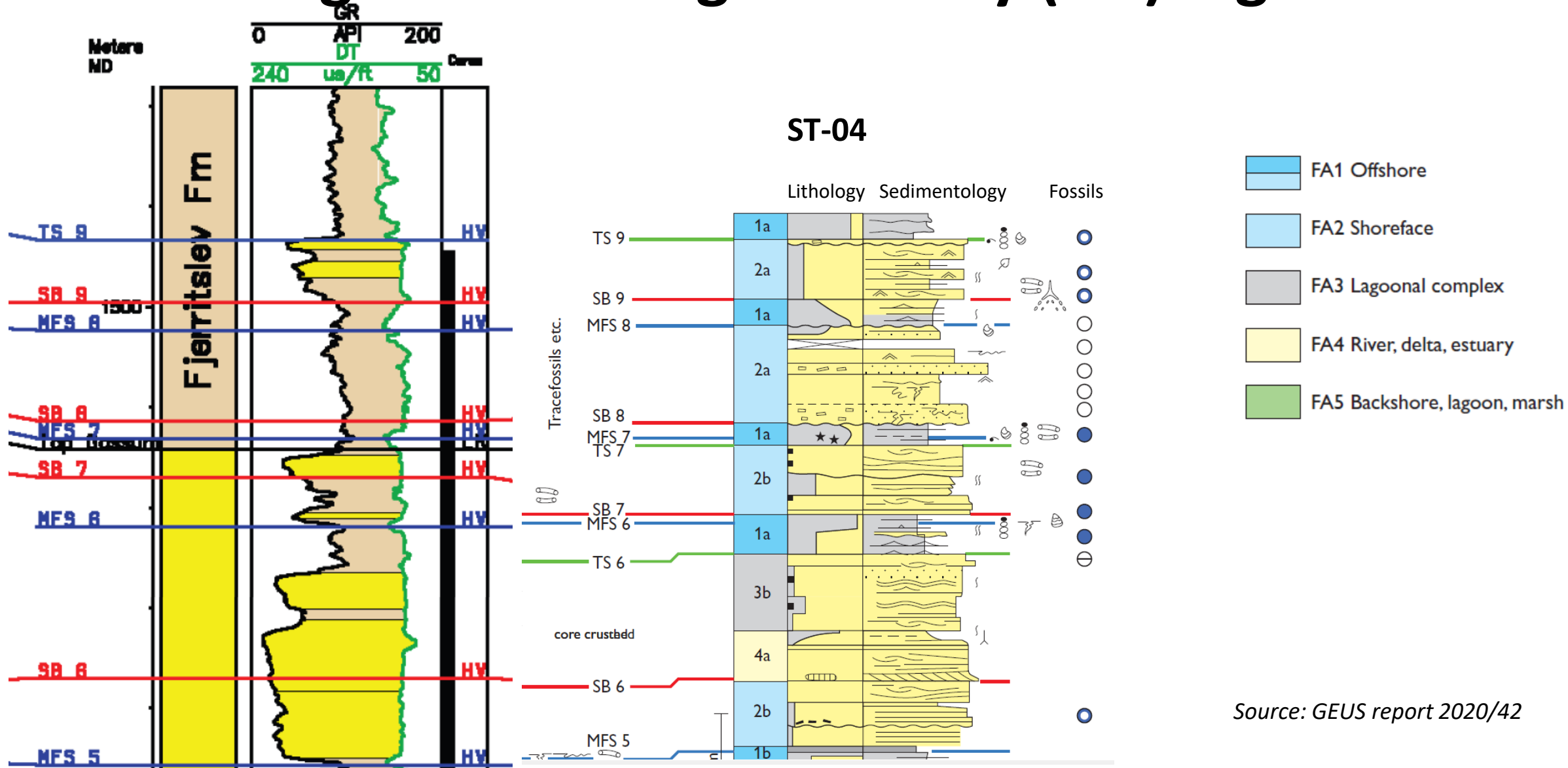
Towards a structural model – Porosity along HS



Towards a structural 3D model



Correlating facies with gamma-ray (GR) logs



Facies description: FA1 – Offshore-shelf

Description	GR signature
FA1 is a broad FA-type dominated by dark grey to black mudstones and storm generated sand-mud heteroliths. The facies range from black laminated claystone (a rare facies), through grey, weakly laminated or structureless mudstone, to dark grey mudstone with streaks of coarse silt to very fine-grained sand. The proximal examples, gradational to FA2 (shoreface), are characterized by sand-dominated tempestites. The mudstones of FA1 normally contain marine palynomorphs; however, some intervals are strongly dominated by terrestrial microfossils (spores and pollen; e.g., lower part of Fjerritslev Fm;). FA1 is divided into two sub-associations: FA1a: Lower offshore-shelf, and FA1b: Upper offshore. These tentative divisions are broad and include potentially deltaic and embayment variants. Facies association 1a is inferred to be a common FA type in the Fjerritslev Formation particularly above TS9, but it is rarely documented in cores. It forms up to several meters thick, aggradational to upward coarsening successions.	FA1a appearance is distinguishable from the GR log as an abrupt change to steadily high values (above TS9), or basal part of large-scale funnel shaped GR signature upwards from the formation boundary (MFS7).

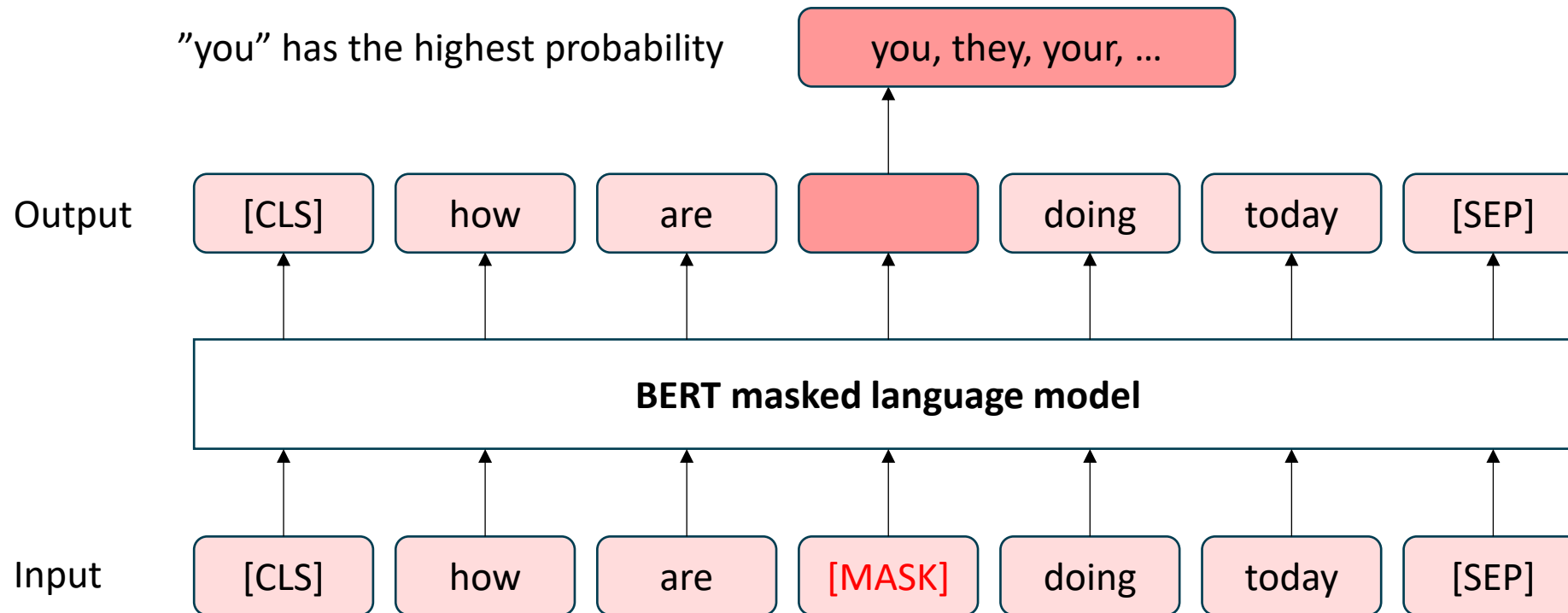


Can a LLM predict GR shape from facies description?

Description	GR signature
FA1a offshore mudstone	Steadily high GR; maximum-GR tops of bell shapes at maximum flooding surfaces.
FA2 shoreface	Sharp-based, aggradational or coarsening-up (funnel) on GR.
FA2a ~FA2b ~ FA4a	All clean sand
FA3a ~ FA1a	Both clay-rich
FA3b tidal	Bell or weak funnel.
FA4 fluvial sandstone	Readily distinguishable from the GR log by its characteristic blocky signature
FA5 palaeosol	Cannot be conclusively recognized from the GR-log alone

Source: GEUS report 2020/42

Large Language Models (LLMs)



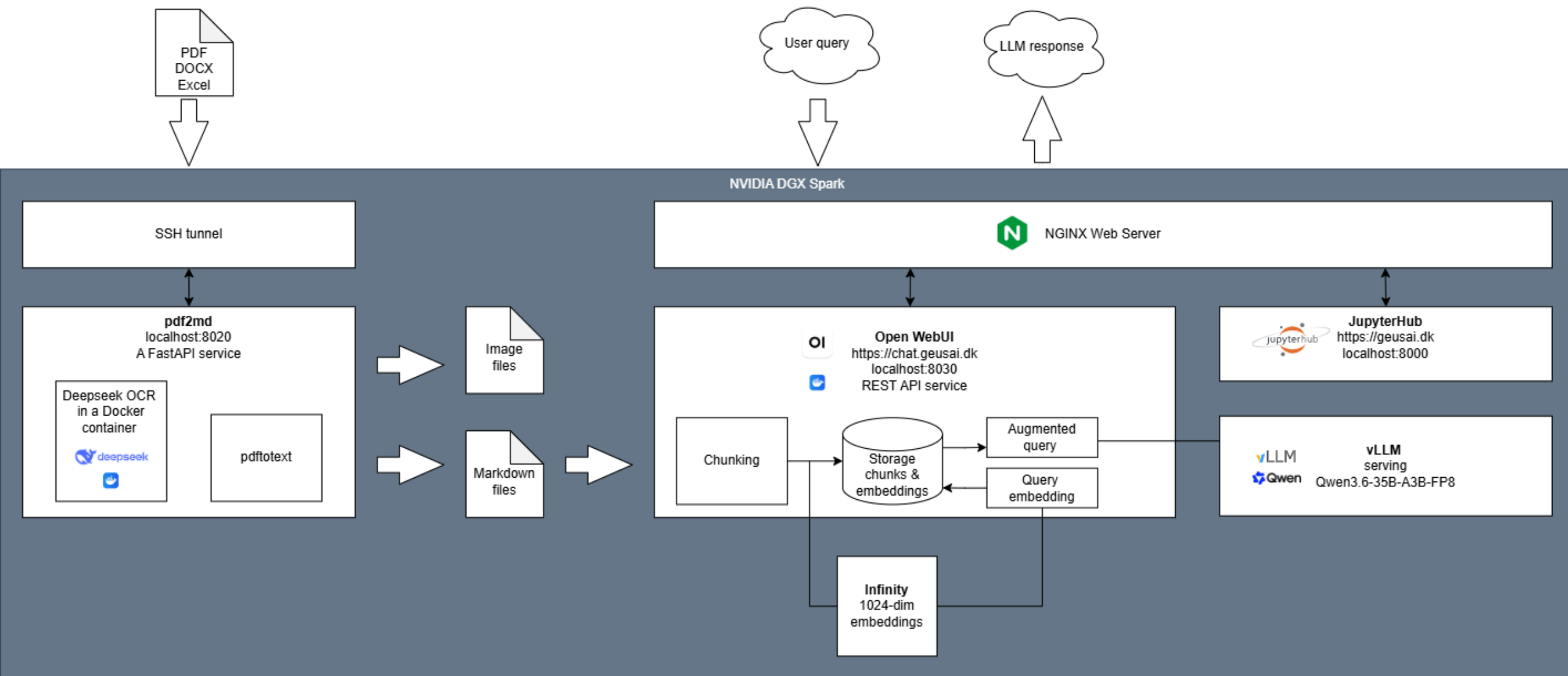
Representation Learning → Next token prediction

Locally hosted AI solution

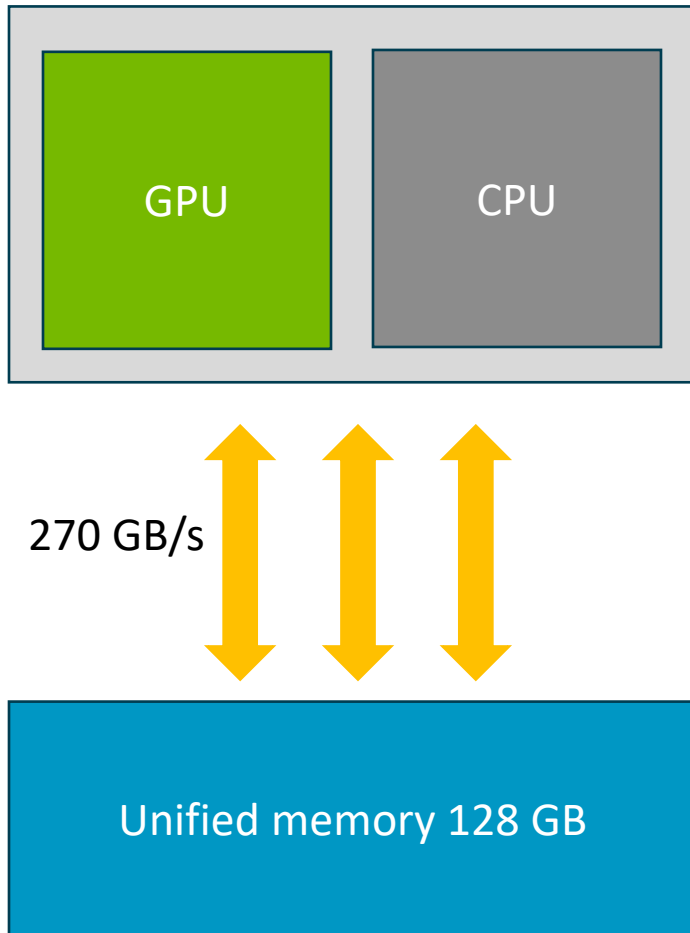
- Cannot use cloud LLM for confidential data
- Hardware borrowed from a Greenland mineral prospecting project
- Fully open-source stack
- Evaluate several open-source LLMs
- Fully sovereign solution at ~50 token/sec (humans read at 5-7 token/sec)



Data ingestion & RAG inference workflows



Towards an optimal LLM configuration



BF16/FP16: 2 bytes per parameter
FP8: 1 byte per parameter
INT4: ~0.5 bytes per parameter

Considerable improvement in token generation speed

- Quantization (FP8/INT8)
- Mixture-of-Experts architecture (active/total params)

Model	Active bytes/token	Generation speed
Gemma-3-27B	14 GB (using INT4 / BF16)	12 tok/sec
Qwen3.6-35B-A3B-FP8	3 GB	50 tok/sec

Model	Total parameters	BF16	FP8
Gemma-3-27B	27B	54 GB	27 GB
Qwen3.6-35B-A3B	35B	70 GB	35 GB

Conclusions & Outlook

- A preliminary conceptual structural geological model built using available data
- Data gap 200–1600 m: only GR in few wells, no core
- Can we utilize general geological knowledge and available GR logs in this range to infer rock properties?
- Sovereign LLM running at conversational speed
- Work with sedimentologists to create a training dataset
- Project dissemination: <https://omgccc.dk> up and running